different internal states in the Mistral-7B (Jiang et al., 2023) in § 5.

4.2 Baselines

To explore query-only uncertainty estimation, we involve straightforward prompt-based approaches as baselines.

Zero-shot Prompt We directly ask the LLM whether it can accurately respond to the query via the following prompt: `"Query: {Query}\n\nAre you capable of providing an accurate response to the query given above? Respond only to this question with 'yes' or 'no' and do not address the content of the query itself."`

In-Context-Learning (ICL) Prompt We ask the LLM whether it can accurately respond to the query and give some examples: `"Are you capable of providing an accurate response to the following query? Respond only to this question with 'yes' or 'no' and do not address the content of the query itself.\n\nQuery: {Example Query 0}\nAnswer: no\n\nQuery: {Example Query 1}\nAnswer: yes...\n\nQuery: {Query}\nAnswer:"`

Perplexity (PPL) Considering the prompt-based methods only use the model's inner knowledge, we also incorporate the distribution of the training dataset and employ a Perplexity (PPL)-based baseline. Assume LLMs are trained on a hypothetical large dataset that perfectly contains every possible query-response pair, where the responses are guaranteed to be faithful. Then, the hallucination estimation can be simply done by checking whether the given query appears in the training corpus (Lee et al., 2021; Kandpal et al., 2023). To determine this threshold, we first calculate the PPL for each query. Subsequently, we identify the optimal PPL threshold that yields the maximum accuracy on our training dataset. This optimal threshold is then applied to the test dataset to gauge the accuracy of our hallucination risk estimation method.

4.3 Estimator Evaluation Protocols

For the classification task with the discrete type predicted, we utilize **F1** and **Accuracy** to measure the quality of predicted categorization.

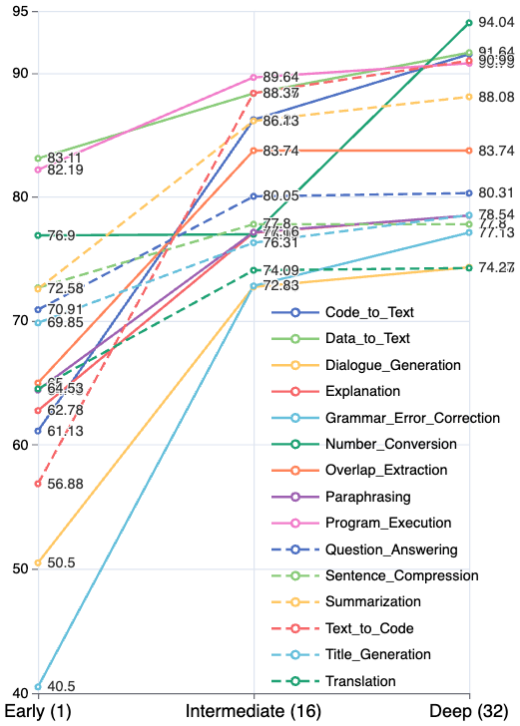


Figure 4: F1 scores of Internal-State from Different Layers for Hallucination Estimation.

Training Task	Testing Task	F1	ACC
QA	Unseen QA	**64.79**	**73.32**
	Translation	51.34	65.10
Translation	Unseen Translation	**74.03**	**73.81**
	QA	20.45	37.50

Table 1: Zero-Shot Automatic Evaluation Results in the Same Task and across Different Tasks.

5 Results and Analysis

5.1 Results for Internal State-based Estimator

(1) Seen/Unseen Query in Training Data We evaluate our internal state-based estimator trained to distinguish unseen and seen questions. The F1 and accuracy scores reach 80.28% and 80.24%. These high results shed light on the effectiveness of our internal state-based method in identifying unseen queries. This phenomenon is aligned with the previous works (Kadavath et al., 2022; Yin et al., 2023b) which find the model can distinguish answerable and unanswerable questions that include future information.

(2) Hallucination Risk faced with the Query For estimating hallucination risk, as depicted in Fig. 3, our methods exhibit superior performance in both F1 and ACC. Notably, its performance remains stable across different tasks. It performs

Task	Internal State	F1	ACC
Dialogue	Llama2	**74.33**	**74.22**
	Mistral	72.39	72.55
QA	Llama2	**82.37**	**82.55**
	Mistral	80.46	81.00
Summarization	Llama2	**88.08**	**88.95**
	Mistral	83.63	85.42
Translation	Llama2	**76.90**	**76.90**
	Mistral	73.10	73.14

Table 2: Automatic Evaluation Results of Internal States from Different Models.

less effectively in the translation task (F1 and ACC 76.90%) while excelling in the Number Conversion task (F1 94.04%, ACC 96.00%). Zero-shot prompt and ICL yield similar results, with ICL slightly outperforming zero-shot prompt. Both methods tend to be overconfident and predict LLM can accurately respond to the query (Recall 99%), which is aligned with the observation of (Xiong et al., 2023). PPL is better than the prompt methods while exhibiting varying performance across tasks. It performs poorly in the translation task (F1 33.73%, ACC 50.36%) but achieves its best performance in the data-to-text task (F1 88.28%, ACC 92.08%).

More results are described in Appendix B including treating separate metrics (Rouge-L, NLI, and QuestEval) as continuous regression labels and different estimator backbones.

5.2 Analysis

Layer Depth *Positively* Correlates with its Prediction Performance. We systematically dissect the contribution of each layer to the overall hallucination risk estimation. We hypothesize that certain layers may be more indicative of hallucinatory propensities than others, and our analysis seeks to validate this hypothesis. As shown in Fig. 4, early Layers perform poorly since they often capture basic syntactic information. Intermediate layers perform better since these layers typically encode more complex semantic relationships. Deep layers perform best and learn hallucination patterns with high-level presentation. This observation is different from Azaria and Mitchell (2023) where middle-layer hidden states of statements perform best in recognizing lying.

Consistency of Internal States across Different LLMs To evaluate the impact of the LLM's Internal State, we use Mistral-7B's internal state to assess Llama2's hallucination risk. As shown in Tab. 2, the results for four common NLG tasks exhibit a decrease compared to Llama2's own internal states. Since different LLMs share a similarity in model architecture and data, there is a potential for zero-shot transfer. Nonetheless, the most effective predictor of LLM's generative performance is still its own internal state, which underscores the importance of considering model-specific assessments rather than universal ones.

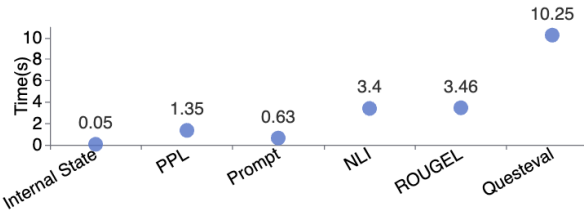


Figure 5: Inference time of various estimation methods.

Internal States Share Features inner-task but do not Cross-task. As shown in Tab. 1, we evaluate the generalization across different NLG tasks and within the same NLG task. Specifically, we examine zero-shot performance in QA and translation. While the zero-shot performance within these individual tasks is acceptable, the cross-task generalization remains relatively weak, aligned with the findings reported by Kadavath et al. (2022).

In addition, we evaluate our estimator trained in QA on the out-of-domain hallucination QA dataset **ANAH** (Ji et al., 2024) to test our estimator's performance in the hallucination aspect and its generalization. ANAH is a bilingual dataset that offers analytical annotation of Hallucinations in LLMs within Generative QA. Our work uses English samples and treats the hallucination type as the label in the testing stage. The F1 score reaches 78.56% and the accuracy is 78.83%. These relatively high results shed light on the effectiveness of our internal state estimator in handling hallucination challenges and further show our generalization capabilities. Therefore, the features in internal states are shared with OOD data within the same task but not shared across tasks.

Internal State as an *Efficient* Hallucination Estimator Our estimator has three linear layers which requires minimal computing power. As shown in Fig. 5, our estimator demonstrates impressive efficiency. Specifically, likelihood-based costs 1.36s per sample, while internal-state-based costs only 0.05s per sample. This rapid inference

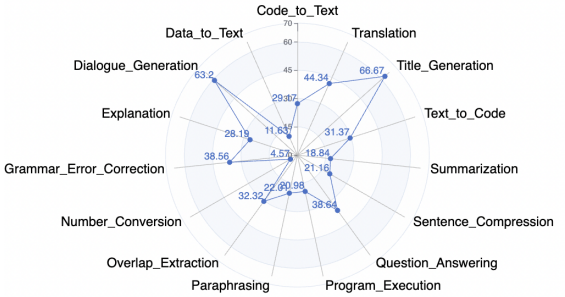


Figure 6: Hallucination Rate for each NLG Task.

speed is essential for real-world applications. The generation time, with a maximum token length of 50 and a batch size of 1, is 3.37 seconds. Notably, Questeval costs the most time 10.25s in total.

Hallucination Rate During the labeling process mentioned in § 3.2, we obtain the hallucination rate in the responses of each task. As illustrated in Fig. 6, the hallucination rate fluctuates significantly across NLG tasks. Among them, Title Generation exhibits the highest rate since its divergent nature and there is no unique and standard answer. In contrast, Number Conversion gains the lowest rate since the task is relatively easy and the answer is fixed leaving less room for hallucination.

Visualizing Tokens Triggering Hallucination To further understand the mechanisms behind hallucination, we dissect the process of the queries triggering hallucinations at a fine-grain level. Inspired by the Gradient-weighted Class Activation Mapping (Grad-CAM) technique (Selvaraju et al., 2017), we quantify the average gradients of input embedding associated with each token in the joint operation of the LLM and estimator. Specifically, we focused on how these tokens influence the LLM's internal state and the subsequent estimation of hallucinations based on this internal state.

Fig. 7 indicates that tokens within an unknown query contribute unequally to the occurrence of hallucinations. We observe that the tokens that are part of unfamiliar named entities or carry critical information exhibit a higher impact. For instance, "*amniotes*" in the QA task and "麻雀" in the translation task gain higher gradients and significantly impact hallucination estimation. This could be attributed to the system's attempts to generate fluent responses despite gaps in its understanding or knowledge about these entities.

Error Analysis Although our method performs better than the baselines in the estimation task, it still generates a few cases of failure. To gain more insight into our model, we present a failure example in Tab. 3 and conduct an error analysis. For example, the estimator predicts that LLM can correctly answer the query "*What is the term for tough, flexible connective tissue that contains the protein collagen?*" But LLM replies "*ligaments*" instead of "*cartilage*", which is hallucinated. the estimator predicts that LLM will hallucinate when faced with "*what appropriately nicknamed pacific location?*" But LLM replies without hallucination.

Question:
How many years ago did early amniotes diverge into two groups ?

Golden Answer:
320 million years

Generated Reply:
…occurred around 360-380 million years ago…

Translation Input:
麻雀说服Tom用魔法指南针找到箱子。

Golden Answer:
The sparrow convinced Tom to use a magic compass to find the box.

Generated Reply:
Maggie convinced Tom to use a magic compass to find the box.

Figure 7: Visualization of Token Contributions to hallucinations in unknown queries for QA (top) and translation (bottom) tasks. Deeper background color means higher contributions and the hallucinated content in the generated reply is marked in pink.

6 Related Work

Knowledge Boundary Researchers investigate the boundary of parametric knowledge in LLMs which aim to uncover *what models know and where their capabilities end*. Current research on the knowledge boundary predominantly narrows down this investigation to specialized tasks, charting a map of knowns and unknowns within these defined borders. In the representative QA field, Rajpurkar et al. (2018) and Yin et al. (2023b) collect datasets containing known and unknown questions and develop classifiers to differentiate them. The concept of unanswerability in these works is universal and *model-agnostic*, such as philosophical questions and unsolved mysteries. In contrast, Kadavath et al. (2022); Slobodkin et al. (2023), and Gottesman and Geva (2024) focus on *model-specific* prediction for questions. The former focuses on whether

Query	Generated Reply	Golden Answer	Trainig Label	Predicted Label
Given a scientific question, generate a correct answer to it. What is the term for tough, flexible connective tissue that contains the protein collagen?	Ligaments are fibrous connective tissue structures that connect bones to other bones and provide support and stability to the joint...	cartilage	hallucinated	faithful
Given a scientific question, generate a correct answer to it. Since 1900, four of the five earthquakes of the greatest magnitude occurred near what appropriately nicknamed pacific location?	Since 1900, four of the five earthquakes of the greatest magnitude have occurred near the Pacific location of the Ring of Fire. The Ring of Fire is an area of...	ring of fire	faithful	hallucinated

Table 3: Negative Samples. The hallucinated context is marked in pink.

the model will answer correctly, while the latter focuses on whether the model linguistically refuses to answer. Out-of-domain or out-of-distribution detection (Zhou et al., 2023; Ryu et al., 2018; Tan et al., 2019; Yang et al., 2021; Zheng et al., 2020) are also relevant areas dealing with the differentiation of unknown/unseen from training data, with main focus on classification tasks. Our method is versatile across various NLG tasks without requiring fine-tuning of LLMs.

Hallucination Detection The phenomenon of hallucination in NLG encourages a variety of detection methods (Min et al., 2023; Ji et al., 2024; Li et al., 2023; Scialom et al., 2021). Some of these methods delve into the internal states for detection. Azaria and Mitchell (2023), for example, collect a true-false statement dataset with *artificial guidance* and the classification results indicate that the LLMs' internal state can reveal the truthfulness of statements. INSIDE (Chen et al., 2024) also leverages LLMs' internal states and proposes EigenScore for evaluating the self-consistency of responses, thereby serving as a proxy for hallucination levels. MIND (Su et al., 2024), an unsupervised training approach, distinguishes the hallucinated continuation text from the original Wikipedia content based on internal states. Snyder et al. (2023) explore the query-only detection within the QA task based on internal states, gradients, and probabilities. On the other hand, Xiao and Wang (2021) shows evidence that higher uncertainty corresponds to a higher hallucination probability. Uncertainty estimation methods, such as (Xiao et al., 2022; Xiong et al., 2023; Kadavath et al., 2022), predict the reliability of their natural language outputs and can also serve as a tool for hallucination detection. Previous works leverage the LLM's internal states for the text to be measured which is not necessary from the same LLM. Differently, this work focuses on self-awareness corresponding to the queries across multiple NLG tasks.

7 Conclusion

Inspired by human self-awareness, this work demonstrates the latent capacity of LLMs to self-assess and estimate hallucination risks prior to response generation. We conduct a comprehensive analysis of the internal states of LLMs both in terms of training data sources and across 15 NLG tasks with over 700 datasets. Employing a probing estimator on the internal states associated with the queries, we assess their self-awareness and ability to indicate uncertainty in two aspects: (1) recognizing whether they have seen the query in training data, achieving an accuracy of 80.28%[5]. (2) recognizing whether they are likely to hallucinate when faced with the query. The results demonstrate that internal state-based self-assessment outperforms PPL-based and prompt-based baselines, with an average estimation accuracy of 84.32% across all tested datasets. In addition, we explore the role of particular neurons in uncertainty and hallucination perception and reveal a positive correlation between the depth of activation layers in an LLM and its predictive accuracy. The consistency of internal states across different models suggests a potential for zero-shot transfer, but model-specific estimation is the optimal strategy. Challenges of generalizing these findings across different tasks are noted, despite observing promising generalizations within the same NLG tasks.

For future work, we aim to refine our methodology to enhance the robustness and generalization across various NLG tasks in the field of hallucination risk assessment. In addition, we will involve a

[5]Please refer to the first parts of § 3.2 and § 5.1.